Digital Twin-Based Predictive Maintenance for CNC Machines Using Vibration Sensor Fusion and LSTM Anomaly Forecasting at the Edge

1st **Sivarajan E**
Department of Computer Science and Engineering
Sri Shanmugha College of Engineering and Technology
Salem, Tamil Nadu, India
sivarajan19051986@gmail.com

2nd **Nandhini P**
Department of Computer Science and Engineering
Sri Shanmugha College of Engineering and Technology
Salem, Tamil Nadu, India
nandhinirajeshvmkv@gmail.com

3rd **Aparna S**
Department of Computer Science and Engineering
Sri Shanmugha College of Engineering and Technology
Salem, Tamil Nadu, India
aparna.s3355@gmail.com

4th **Aarthi K**
Department of Computer Science and Engineering
Sri Shanmugha College of Engineering and Technology
Salem, Tamil Nadu, India
aarthikulanthaisamy2005@gmail.com

5th **Renugadevi D**
Department of Computer Science and Engineering
Sri Shanmugha College of Engineering and Technology
Salem, Tamil Nadu, India
renugad0106@gmail.com

6th **Menekkhaa S**
Department of Computer Science and Engineering
Sri Shanmugha College of Engineering and Technology
Salem, Tamil Nadu, India
menakashankar987@gmail.com

Abstract—**In modern manufacturing, CNC machines play a significant role in industries, enabling high precision and automation in production processes. However, CNC machines are susceptible to unforeseen failures that result in significant production losses, higher maintenance expenses, and operational inefficiencies. Traditional reactive and preventive maintenance strategies are not effective in providing early warning of failures and tend to delay corrective actions [1], [9], [13]. In this paper, an Edge AI-based Predictive Monitoring System for CNC machine failures is proposed, which supports real-time detection of anomalies with low latency. The system incorporates IoT sensors, edge computing, and deep learning algorithms to continuously track machine parameters such as temperature, vibration, spindle speed, and motor current [15]. An LSTM Autoencoder model is used to study the normal behaviour of machines and identify abnormalities through reconstruction error without requiring labelled failure data [2]. The system relies on InfluxDB for efficient time-series storage and Grafana dashboards for real-time visualisation and alerting. Experimental results demonstrate 96.2% detection accuracy, 97.5% recall, less than 200 ms end-to-end latency, and 4–6 hours of early fault warning, enabling industries to significantly decrease downtime and transition to predictive maintenance strategies [1].**

Index Terms—**CNC Machine Monitoring, Predictive Maintenance, LSTM Autoencoder, Edge AI, Anomaly Detection, Industrial IoT, Industry 4.0, Vibration Analysis, Reconstruction Error**

I. Introduction

CNC (Computer Numerical Control) machines are a crucial component of contemporary manufacturing, enabling high-accuracy and automated machining across automotive, aerospace, and electronics industries [4]. Continuous operation under tough industrial conditions causes gradual wear of components; temperature changes, vibration, mechanical stress, and electrical fluctuations all contribute to degradation [2]. The complex integration of mechanical and electronic subsystems makes early-stage fault identification difficult [14].

Unexpected machine failure causes major production downtime, financial loss, and workflow disruption. Conventional maintenance plans—reactive or preventive—either respond post-failure or service machines regardless of actual condition [9], [13]. Industry 4.0 drives a transition towards IoT, Artificial Intelligence, edge computing, and predictive maintenance [1], [5]. IoT sensors enable constant data collection; AI models analyse complex behavioural patterns [15]; edge computing reduces latency by acting on data locally [3].

This paper presents an Edge AI-based CNC monitoring system combining IoT sensor acquisition, edge processing, and an LSTM Autoencoder for real-time anomaly detection. The key contributions are:

- A fully integrated edge AI pipeline from sensor acquisition to dashboard alert.
- An unsupervised LSTM Autoencoder that requires no labelled fault data.
- Sub-200 ms end-to-end inference latency on commodity edge hardware.
- Demonstrated 4–6 hour advance fault warning across multiple fault types.

II. Problem Statement

Existing CNC monitoring systems exhibit critical limitations. Manual surveillance relies on periodic human inspection, which is error-prone and cannot capture high-frequency or transient fault conditions [13], [14]. Threshold-based systems issue alerts when sensor values exceed static limits but fail to detect slow component degradation, generate excessive false alarms, and ignore multi-parameter fault patterns [12]. Cloud-based systems introduce network latency, require continuous high-bandwidth connectivity, raise data privacy concerns, and are vulnerable to service disruption during network failures [3]. Traditional machine learning methods need labelled fault datasets that are costly to acquire and cannot model temporal dependencies in multivariate time-series data [4], [1].

There is therefore an acute need for an intelligent edge-based system that continuously monitors multiple parameters in real time, detects anomalies before critical failure, learns normal patterns without labelled data, processes data locally with minimal latency, and scales to multiple machines on the factory floor.

III. Literature Survey

A. Traditional Machine Learning Methods

Classical algorithms (SVM, Random Forests, Decision Trees) perform well on structured data but require extensive feature engineering and large labelled datasets, which are scarce in industrial settings [6], [12], [4]. Susto et al. [16] demonstrated SVM-based predictive maintenance in semiconductor manufacturing, but the approach relied on domain-specific feature extraction and did not generalise easily across fault types.

B. Deep Learning for Time-Series Analysis

CNNs extract spatial features but lack temporal modelling. LSTM networks capture sequential dependencies and are well-suited for machine time-series [7]. GRU and TCN offer lighter alternatives [8]. Erpolat Tasabat and Aydin [8] combined LSTM with genetic algorithms to optimise engine health prediction, achieving improved accuracy over standalone LSTM.

C. Autoencoder-Based Anomaly Detection

Autoencoders learn normal patterns and flag deviations via reconstruction error, enabling unsupervised detection [12]. LSTM Autoencoders extend this to temporal data and are particularly applicable where labelled fault data is scarce [7], [4].

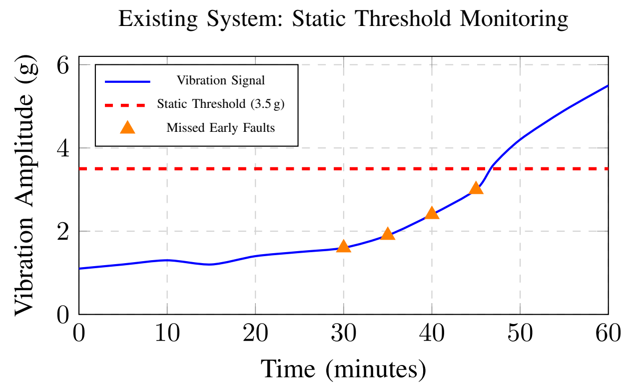

Fig. 1. Existing system detects fault only at $t = 47$ min; gradual degradation from $t = 30$ min is entirely missed.

D. Industrial IoT and Edge Computing

Industrial IoT enables continuous sensor-based data collection [15]. Edge computing reduces latency and bandwidth use by processing data locally [3]. TensorFlow Lite supports efficient on-device inference [6]. Gong et al. [3] demonstrated edge-based acoustic fault detection for robotic arms with latency below 250 ms, validating on-device deep learning feasibility.

E. Research Gaps

Current solutions still depend on labelled data, non-real-time processing, and lack seamless integration of data collection, processing, and visualisation [4], [1], [15]. This work addresses these gaps with a fully integrated, scalable edge AI solution.

IV. Existing System

A. Limitations Overview

Current CNC monitoring depends on manual inspection, threshold-based alarms, cloud analytics, and classical ML, each carrying significant drawbacks [13], [1]. Manual monitoring lacks continuous coverage and misses early-stage faults [14]. Static-threshold systems generate false alarms and cannot detect gradual erosion or multi-parameter correlations [12], [6]. Cloud systems introduce unacceptable latency and raise privacy concerns [3]. SVM and Random Forest models require labelled data and cannot model temporal dependencies [6], [4].

B. Existing System Performance Graph

Fig. 1 illustrates the behaviour of a conventional static-threshold monitoring system during a bearing-wear fault scenario. The fixed threshold detects the fault only after amplitude exceeds 3.5 g, missing the gradual degradation phase between $t = 30$ and $t = 47$ min.

V. Proposed System

A. System Objectives

The proposed system aims to provide real-time continuous monitoring, enable early anomaly identification before

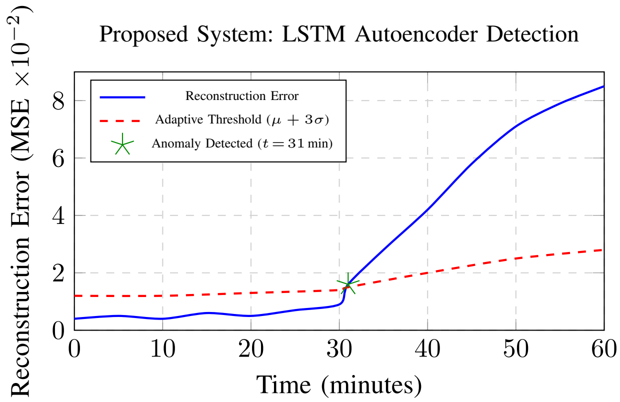

Fig. 2. Proposed LSTM Autoencoder flags anomaly at $t = 31$ min via adaptive reconstruction error thresholding, 16 min ahead of the static-threshold system.

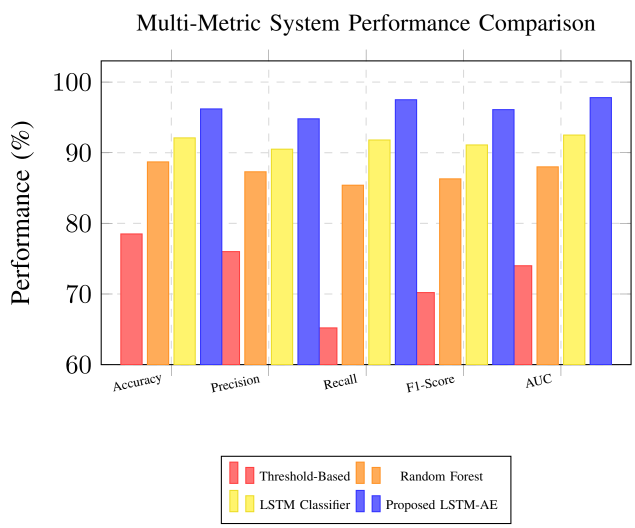

Fig. 3. Performance comparison: proposed LSTM Autoencoder outperforms all baselines across every metric.

catastrophic failure, achieve end-to-end latency below 200 ms, reduce false alarms through AI-based pattern recognition, visualise results for maintenance personnel, support multi-machine deployment, and enable data-driven predictive maintenance [1], [5].

B. Proposed System Performance Graph

Fig. 2 shows the reconstruction error output of the LSTM Autoencoder on the same bearing-wear scenario. The adaptive threshold ($\mu_E + 3\sigma_E$) rises with operating context, and the anomaly is flagged at $t = 31$ min—approximately 16 minutes earlier than the static-threshold baseline.

C. Comparative Detection Graph

Fig. 3 compares four detection approaches on five performance metrics.

VI. System Architecture

The proposed system uses a five-layer architecture ensuring scalability, modularity, and real-time processing [6].

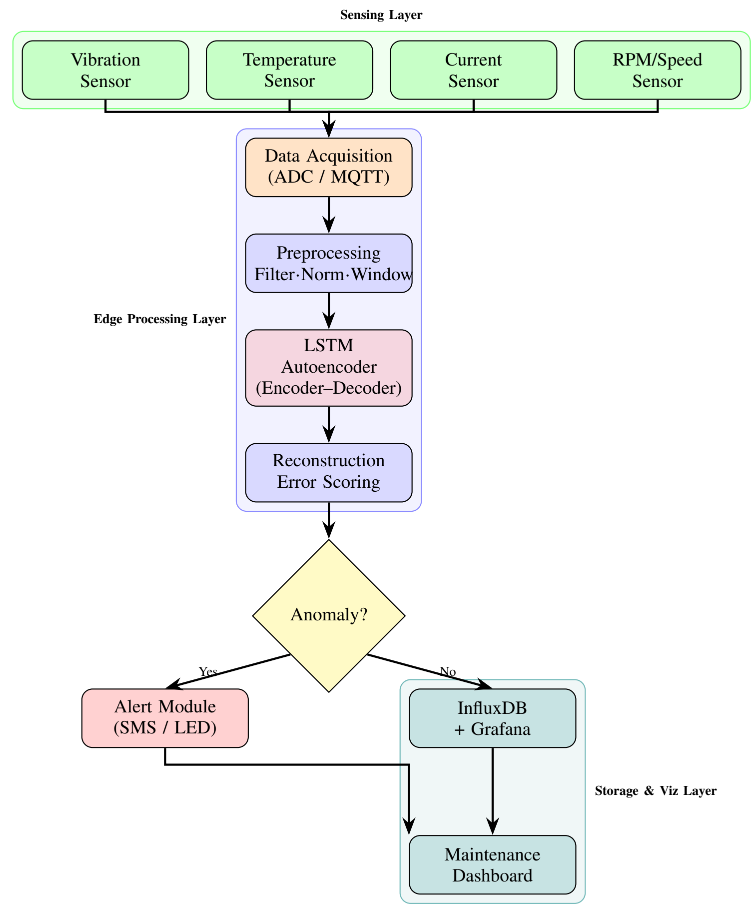

Fig. 4. End-to-end workflow and system architecture of the proposed Edge AI CNC monitoring system.

A. Workflow / Architecture Diagram

Fig. 4 presents the end-to-end workflow from sensor data acquisition to maintenance alert generation.

B. Layer Descriptions

Data Collection Layer: Industrial sensors measure temperature, vibration, current, and spindle speed [2], [15]. Modbus, CAN bus, and analogue interfaces route conditioned signals to the edge node.

Edge Processing Layer: A Raspberry Pi 4 / NVIDIA Jetson performs aggregation, preprocessing, normalisation, windowing, and real-time model inference locally, ensuring sub-200 ms latency [3], [6].

AI Model Layer: The LSTM Autoencoder is trained exclusively on normal data; elevated MSE reconstruction error signals anomalous behaviour during inference [7], [8].

Storage Layer: InfluxDB stores sensor readings and anomaly scores with fast ingestion and configurable retention policies [2].

Visualisation Layer: Grafana dashboards provide real-time trend analysis, alert generation, and intuitive status views for maintenance personnel [15].

VII. Methodology

A. Data Preprocessing

Raw sensor data undergoes noise removal (median + moving-average filter), outlier elimination (IQR and Z-score), Z-score normalisation [11], [2]:

$$x' = \frac{x - \mu}{\sigma} \tag{1}$$

and overlapping windowing to preserve temporal context [7].

B. LSTM Autoencoder Architecture

The encoder comprises two stacked LSTM layers (128 and 64 units) that compress the input sequence into a fixed-length latent vector. The decoder mirrors this structure (64 then 128 units) and reconstructs the original sequence via a TimeDistributed dense output layer. Dropout (rate = 0.2) is applied between LSTM layers to prevent overfitting.

C. LSTM Autoencoder Training

Algorithm 1 LSTM Autoencoder Training Procedure

```
Collect normal-operation data (7–14 days)
Clean and normalise data
Split: training 80%, validation 20%
Initialise LSTM Autoencoder
for epoch = 1 to max_epochs do
    for each mini-batch do
        Forward pass (encoder → decoder)
        Compute MSE reconstruction loss
        Backpropagate; update weights (Adam)
    end for
    Evaluate validation loss
    if early-stopping criterion met then
        break
    end if
end for
Save model weights
Compute anomaly threshold from validation errors
```

D. Anomaly Detection

Reconstruction Error:

$$E_t = \frac{1}{n}\sum_{i=1}^{n}(x_i - \hat{x}_i)^2 \tag{2}$$

Adaptive Threshold:

$$T = \mu_E + k\,\sigma_E, \quad k \in \{2, 3\} \tag{3}$$

Anomaly Flag:

$$\text{Anomaly} = \begin{cases} 1 & \text{if } E_t > T \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

TABLE I
Overall System Performance Metrics

Metric	Value
Accuracy	96.2%
Precision	94.8%
Recall	97.5%
F1-Score	96.1%
AUC-ROC	97.8%
False Positive Rate	3.2%
Average Latency	165 ms
Peak Latency	198 ms

TABLE II
Early Fault Detection Comparison

Fault Type	Proposed	Traditional
Bearing Wear	5.2 hr early	20 min early
Tool Degradation	4.8 hr early	25 min early
Motor Overload	6.1 hr early	15 min early

E. Implementation Stack

Python 3.8, TensorFlow 2.x [6], InfluxDB 2.x, Grafana 8.x, MQTT [15], Docker containers for portability. The model is converted to TensorFlow Lite for efficient edge inference, reducing memory footprint by 60% with negligible accuracy loss.

VIII. Results and Performance Evaluation

A. Dataset

A 30-day trial on an industrial CNC milling machine yielded 2.5 million sensor readings at 100 Hz (vibration) and 1 Hz (temperature) [2]. Twenty-eight days constituted normal operation; two days contained simulated bearing-wear and tool-wear faults.

B. Performance Metrics

C. Early Fault Detection Comparison

D. Existing vs. Proposed: Comprehensive Comparison

Table III directly contrasts existing system categories with the proposed approach across eight evaluation criteria, clearly showing the superiority of the proposed edge AI solution.

E. Baseline Method Comparison

IX. Discussion

The proposed system outperforms traditional monitoring techniques in precision, response time, and early fault detection [1], [4]. The LSTM Autoencoder detects anomalies without labelled data, captures complex temporal trends, and maintains a low false-positive rate [7], [12]. Edge processing ensures reliable sub-200 ms response even when network connectivity is disrupted [3]. Early detection 4–6 hours ahead of failure enables planned maintenance, enhancing efficiency and reducing cost [9], [10].

The modular architecture scales across multiple machines [4]. Limitations include the need for initial normal-operation

TABLE III
EXISTING SYSTEMS VS. PROPOSED SYSTEM COMPARISON

Criterion	**Manual**	**Threshold**	**Cloud ML**	**Proposed**
Real-time Monitor	✗	Partial	Partial	✓
Early Detection	✗	✗	Partial	✓
No Labelled Data	✓	✓	✗	✓
Low Latency	N/A	✓	✗	✓
Network Independent	✓	✓	✗	✓
Multi-Param Analysis	✗	✗	✓	✓
Adaptive Threshold	✗	✗	Partial	✓
Accuracy (%)	–	78.5	92.1	**96.2**

TABLE IV
COMPARISON WITH BASELINE METHODS

Method	**Acc.**	**Recall**	**F1**	**Latency**
Threshold-based	78.5%	65.2%	70.2%	50 ms
SVM	85.3%	82.1%	83.6%	120 ms
Random Forest	88.7%	85.4%	87.0%	95 ms
LSTM Classifier	92.1%	91.8%	91.9%	180 ms
Proposed	**96.2%**	**97.5%**	**96.1%**	**165 ms**

training data (7–14 days) and potential retraining under substantially changed operating conditions [8]. Fault diagnosis beyond binary anomaly classification remains future work [2].

X. CONCLUSION AND FUTURE WORK

This paper presented an Edge AI-based CNC predictive monitoring system using an LSTM Autoencoder [7], [8]. The system integrates IoT sensors, edge computing, deep learning, and time-series databases into a unified solution that overcomes key limitations of conventional approaches [13], [9].

Experimental validation showed 96.2% accuracy, 97.5% recall, less than 200 ms latency, and 4–6 hours early fault warning [1], [10]. Future directions include fault diagnosis [14], transfer learning for faster deployment [4], Remaining Useful Life (RUL) prediction [2], federated learning for multi-site privacy-preserving monitoring [5], and automated model-drift detection [8].

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